Vitamins

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1st Stage

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Introduction:

- Vitamins are a group of organic nutrients, required in small quantities for a variety of biochemical functions that, generally, cannot be synthesized by the human body and must therefore be supplied in the diet.
- In 1912, biochemist Casimir Funk was the first to coin the term “vitamin”.
- The word vitamin comes from the Latin word “vita”, meaning life, and “amine” referring to a nitrogenous substance essential for life.

- Most vitamins function as coenzymes.
- They don’t enter in tissue structure, and are not a source of energy.
- Nutritional value lost by bacteria, light, oxidation and heat.
- The average daily requirement for vitamins differ from one vitamin to another and influenced by many factors such as age, gender, physical stress, pregnancy, lactation, growth, wound healing, ...etc.

Classification of Vitamins

Water soluble vitamins

- Soluble in water
- Easily absorbed
- No requirement of carrier protein
- Have a threshold for urinary excretion.
- Are not stored in body except vitamin B9 & B12.
- Deficiency manifests rapidly as there is no storage.
- Act as co-enzyme
- Examples: B-Complex & Vitamin C
- Most of vitamins can’t be synthesized in the body except, B7 & B9.

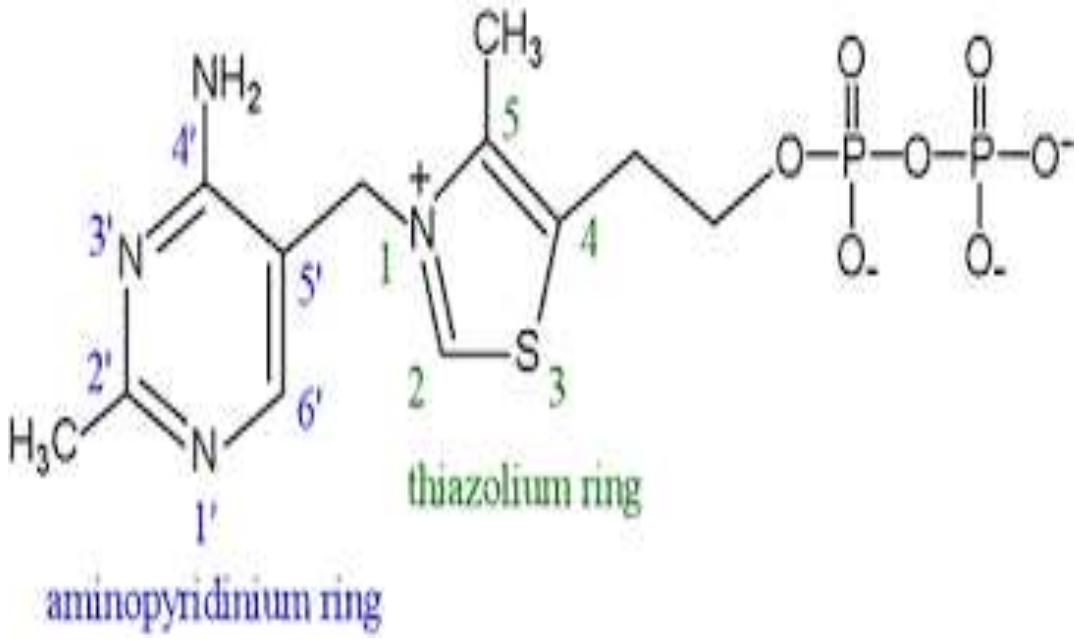
Fat soluble vitamins

- Soluble in fat
- Require bile salts and fat
- Requires carrier proteins
- Are not excreted
- Are generally stored in liver.
- Deficiency manifests only when stores are depleted.
- Do not act as co-enzyme
- Examples: A, D, E, & K.
- Only D & K synthesized in the body.

Water Soluble Vitamins

1- Vitamin B1(Thiamine):

- It is also called aneurin, composed of 2 rings: pyrimidine ring and thiazole ring.
- The biological active form of the vitamin is Thiamine Pyrophosphate (TPP).
- It is formed by transfer of a pyrophosphate group from ATP to thiamine.
- TPP serves as a coenzyme.



- Essential for release of energy from food.
- Necessary for appetite and good health.
- Needed for normal functioning of nervous system.

Recommended Daily Allowance:

1.2 mg	for men
1.1 mg	for women

Sources of B1

Animal products

Organ meats | Fish | Meat

Plant products

Fruits | Nuts | Whole grain cereals

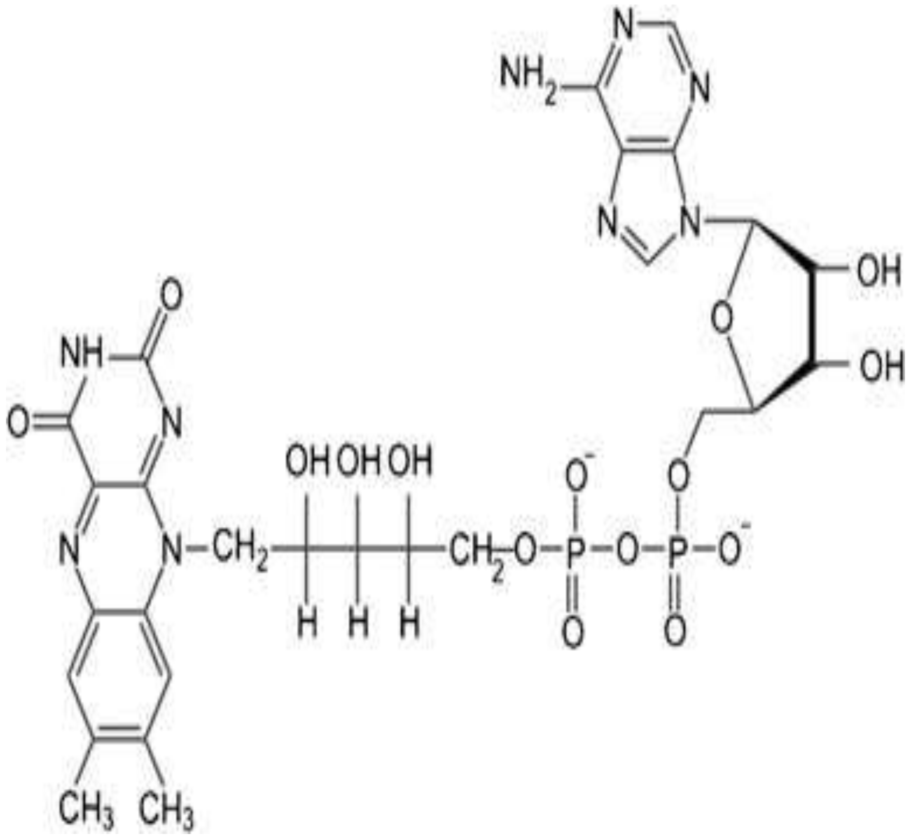
Deficiency of B1:

- Beri-beri disease: This is a severe thiamine-deficiency syndrome found in areas where polished rice is the major component of the diet.

  - It is a neurological disease, characterized by: Dry skin, Irritability, Disorderly thinking, Progressive paralysis.

2- Vitamin B2 (Riboflavin):

- The two biologically active forms are: Flavin mononucleotide(FMN), and Flavin adenine dinucleotide(FAD).
- FAD formed by the transfer of an AMP moiety from ATP to FMN.
- Involved in oxidation-reduction reactions.
- FMN and FAD are involved in metabolism of carbohydrates, proteins and fats.
- Necessary in growth, repair, development of body tissues - healthy skin, eyes and tongue.



Recommended Daily Allowance

1.3 mg	for men
1.1 mg	for women

Sources of B2

Animal products

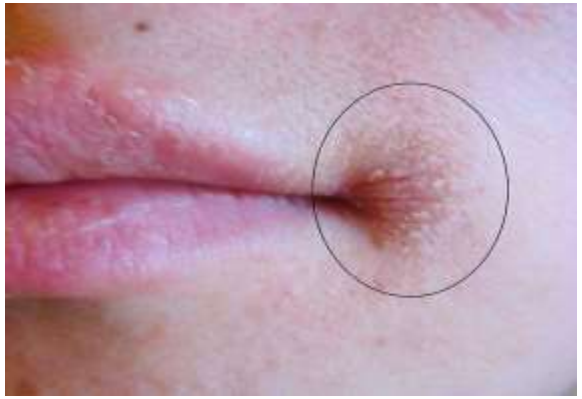
Meat | Milk & Dairy products | Eggs

Plant products

Vegetables | Whole grains | Nuts

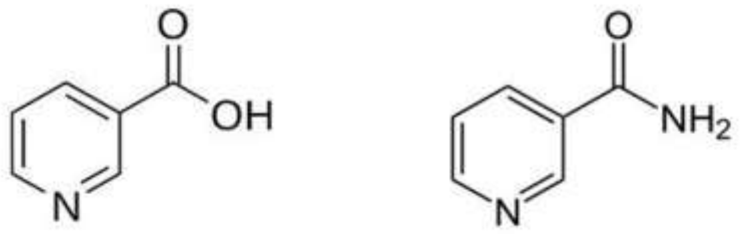
Deficiency of B2:

- cheilosis (painfull fissuring at the corners of the mouth: skin cracks).
- glossitis (the tongue appearing smooth and purplish red).
- dermatitis

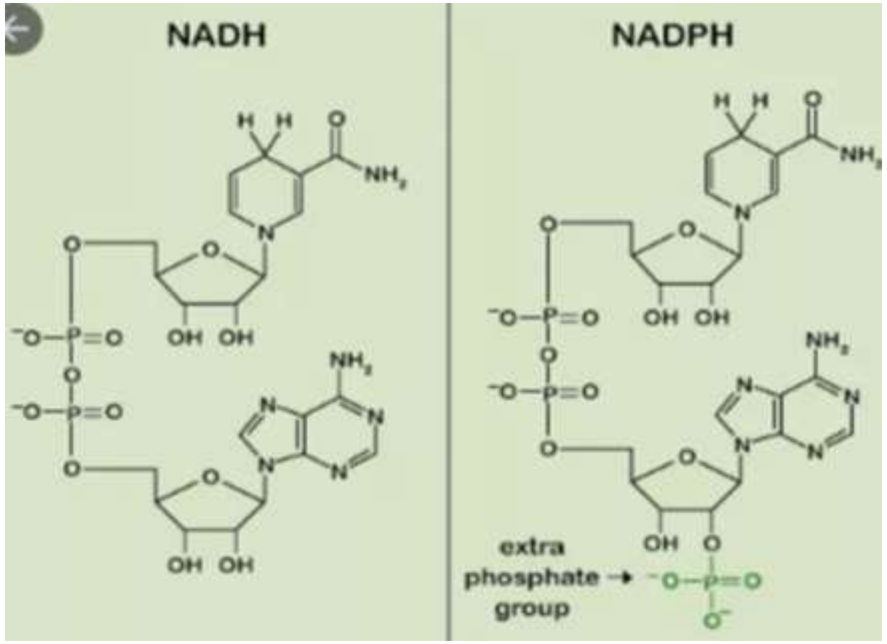


3- <u>Vitamin B3 (Niacin):</u>

- also known as niacin or nicotinic acid.
- Its coenzyme forms are NAD (Nicotinamide adenine dinucleotide) and NADP.
- It is essential for oxidation-reduction reactions.
- It regulates cell division, development and proliferation of tissues.



Niacin Nicotinamide



Recommended Daily Allowance

niacin: 16 mg for men
14mg for women

Sources of B3

Animal products

Meat | Dairy products & eggs | Fish

Plant products

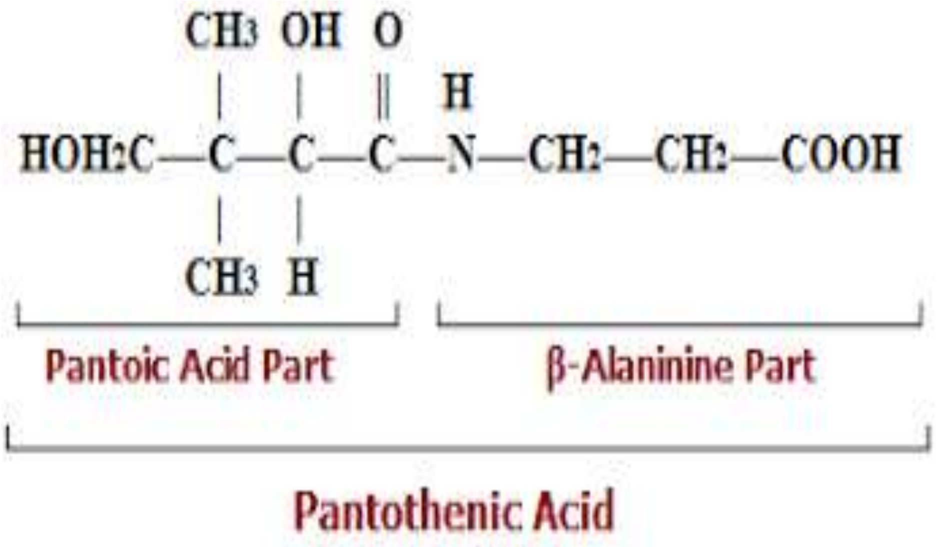
Avocados | Nuts | Whole grains

Deficiency of B3:

- Pellagra, a disease involving the skin, gastrointestinal tract, and CNS. The symptoms of pellagra progress through the three diseases: dermatitis, diarrhea, and dementia, if untreated leads to death.

4- Vitamin B5 (Pantothenic acid):

- The word "Pantothenic Acid" derived from Greek word pantos meaning everywhere.
- It is a component of Coenzyme-A (HS-CoA), a carrier of acyl groups.
- also a component of the acyl carrier protein (ACP), that participates in fatty acid synthesis.



- Functions in the synthesis and degradation of fatty acids, synthesis of steroids (cholesterol), steroid hormones, sphingosine, and citrate.
- It is present in food either free form or coenzyme form.
- Coenzyme form is hydrolyzed to free form by intestinal pyrophosphatase.

Recommended Daily Allowance

5 mg for men & women

Sources of B5

Animal products

Meat | Dairy products & eggs | Fish

Plant products

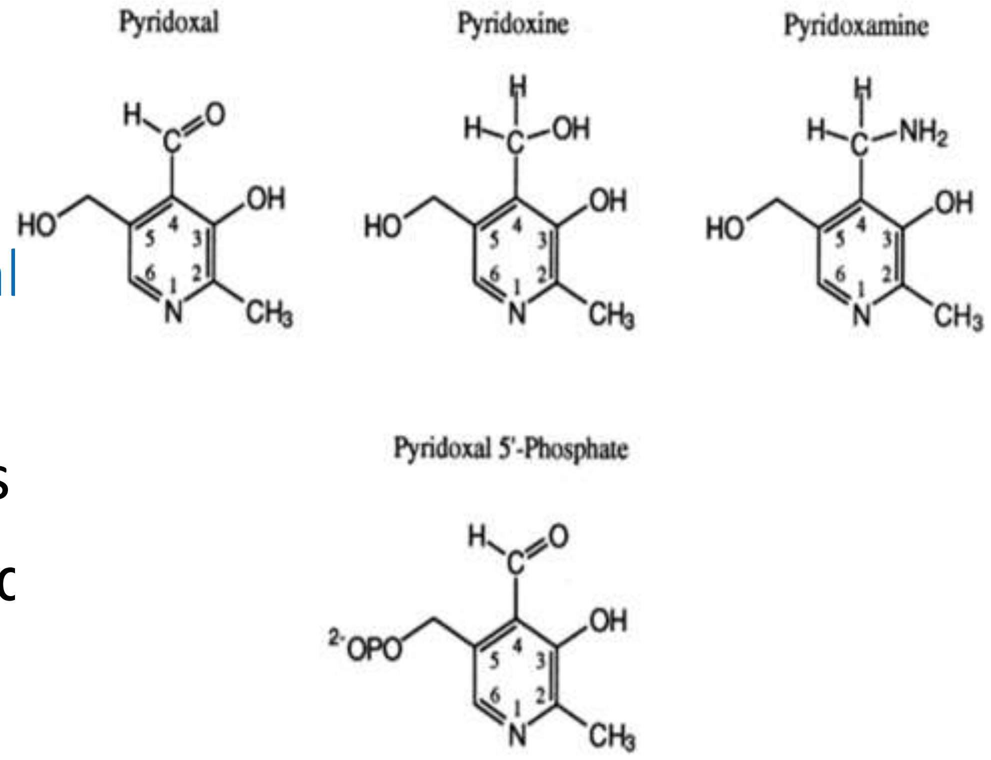
Broccoli | Tomatoes | Mushrooms

Deficiency of B5:

- Paresthesia: sensation of tingling, pricking of skin.
- Anemia
- Insomnia
- Reduced immunity, impaired antibody response.

5- <u>Vitamin B6 (Pyridoxine):</u>

- Vitamin B6 is a collective term for pyridoxine, pyrodoxal, and pyrodoamine (all derivatives of pyridine).
- They differ only in the nature of the functional group attached to the ring.
- Pyridoxine occurs primarily in plants, whereas pyrodoxal and pyrodoxamine are found in fooc obtained from animals.



- All three compounds can serve as precursors of the biologically active coenzyme, piyrodoxal phosphate (PLP).
- The metabolically active form of vitamin B6, is involved in many metabolic process such as amino acid, glucose and lipid metabolism, neurotransmitter synthesis, histamine synthesis, hemoglobin synthesis and function, and gene expression.

Recommended Daily Allowance

1.3 mg for men & women

Sources of B6

Animal products

Liver | Chicken | Fish

Plant products

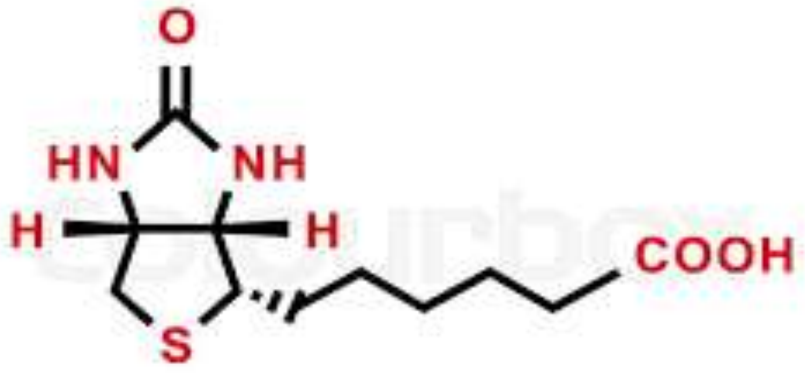
Chickpeas | Nuts| Bananas & oranges

Deficiency of B6:

- Rough skin
- Insomnia (lowered GABA (γ- *amino butiric acid*).
- Anemia, Hb ↓
- ↓ Biogenic amines formation

6- Vitamin B7 (Biotin):

- Vitamin B7 or Biotin, also known as vitamin H or coenzyme R.
- Biotin is a coenzyme for carboxylases, involved in the synthesis of fatty acids, isoleucine, valine, and in gluconeogenesis.
- It is required in very small amounts, because it's available from intestinal bacteria.



VITAMIN B7

- Activates protein/amino acid metabolism in the hair roots (the maintenance of normal hair), and fingernail cells.
- Synthesis urea and purines.
- Involved in the maintenance of normal skin and mucous membranes.
- Maintaining the normal function of the nervous system.
- Raw egg should be avoided, because avidin (egg protein) binds biotin very tightly and may lead to a biotin deficiency (cooking eggs denatures avidin so it does not bind biotin).

- The large intestine microbiota synthesizes amounts of biotin similar to the amount taken in the diet, and a significant portion of this biotin exists in free form easily can be absorbed.

Recommended Daily Allowance

30µg for men & women

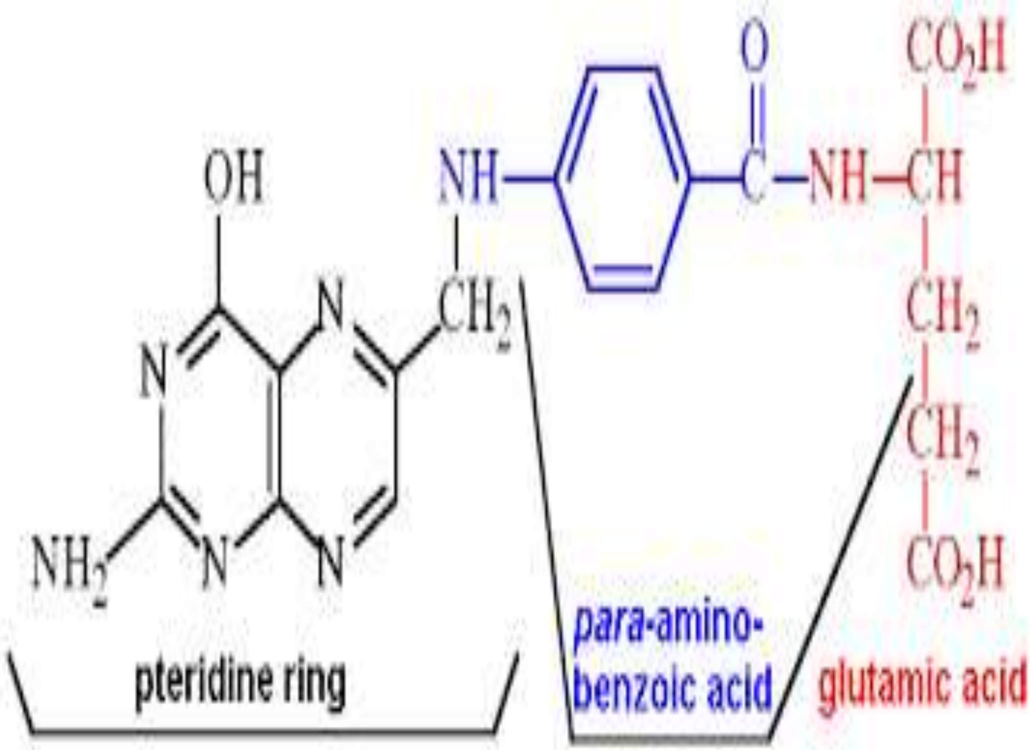
Sources of B7

Deficiency of B7:

- Seborrheic dermatitis, causes scaly patches, red skin and stubborn dandruff.
- Seborrheic dermatitis can also affect oily areas of the body, such as the face, sides of the nose, eyebrows, ears, eyelids and chest.
- Glossitis
- Nausea and loss of appetite.

7- Vitamin B9(Folic acid):

- Vitamin B9 also called folate or folic acid.
- Folic acid is the synthetic form of B9, found in supplements and fortified foods, while folate occurs naturally in foods.
- Tetrahydrofolate (THF or FH4) is the active form of folic acid.



- THF receives one-carbon fragments from donors such as serine, glycine, and histidine and transfers them to intermediates in the synthesis of amino acids, purines, and thymine, a pyrimidine found in DNA.
- Folic acid in combination with vitamin B12 is essential for RBC formation, and maturation.
- Development of brain, spinal cord and skeleton in fetus.

Recommended Daily Allowance

400 µg for men & women

Sources of B9

Animal products

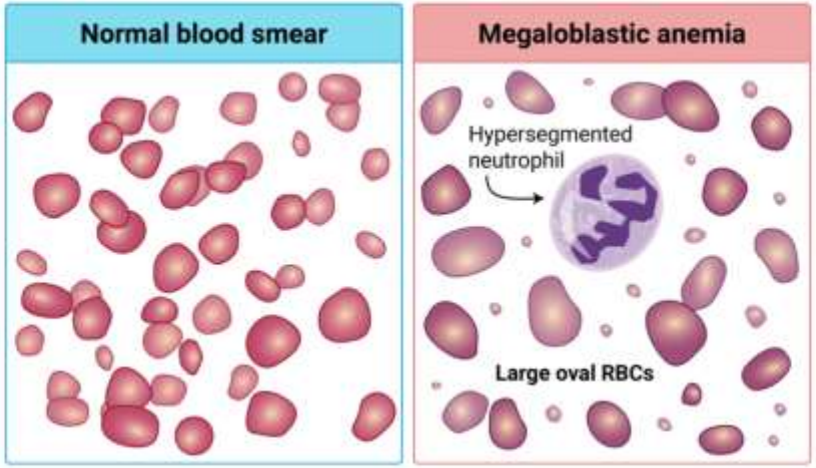
Liver | Dairy products | Egg yolk

Plant products

Oranges | Dark leafy vegetables | Peanuts & beans

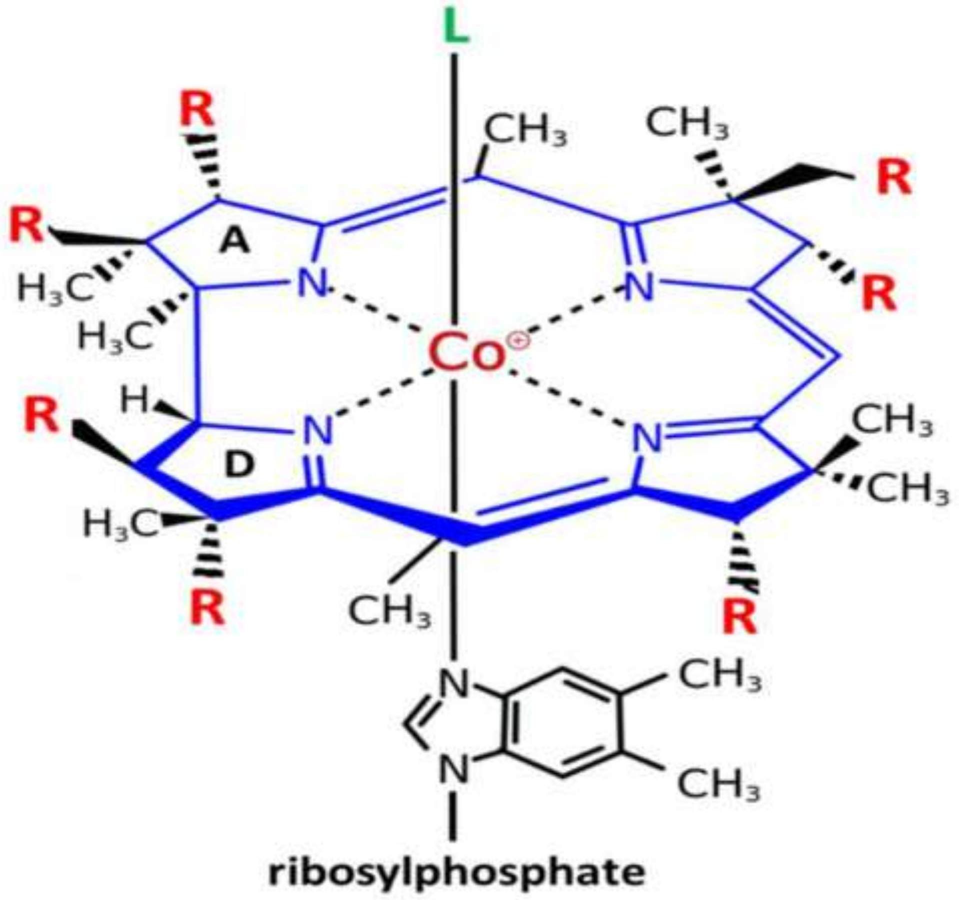
Deficiency of B9:

- Megaloblastic anemia, very large red blood cells.
- Neural tube defect.
- Fatigue in mild cases.
- Glossitis
- Growth failure
- Gastrointestinal Tract Disturbance.



8- Vitamin B12 (Cobalamine):

- Vitamin B12 is a complex compound called cobalamin, it is cobalt containing porphyrin.
- It's found only in animal products, also synthesized in the human large intestine by microorganisms, but with no physiological or nutritional benefit for humans.
- It requires to IF for it's absorption.
- In fact, vitamin B12 exists in four very similar chemical forms, as Methylcobalamin, Adenosylcobalamin, Cyanocobalamin, and Hydroxycobalamin.



- A key role in the normal functioning of the brain and nervous system via the synthesis of myelin (myelinogenesis).
- It is involved in the metabolism of every cell of the human body, especially affecting DNA synthesis, fatty acid and amino acid metabolism.
- Formation of red blood cells.
- Vitamin B12 in food is usually in coenzyme form and bound to proteins in diet.
- Liver is the principal storage site of vitamin B12, other tissues like kidney, heart and brain also store the vitamin, but in lesser amounts.

Recommended Daily Allowance

2.4 µg for men & women

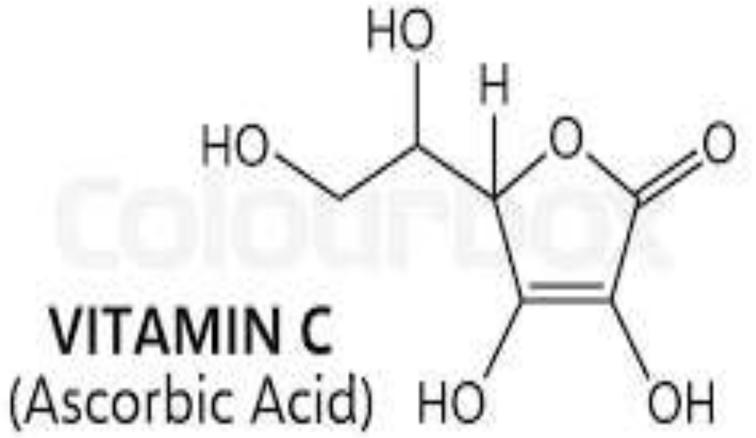
Sources of B12

Deficiency of B12:

- Pernicious anemia, and Megaloblastic anemia.
- Neurological disorders; memory loss, depression, numbness.
- Shortness of breath.
- Fatigue.

9- Vitamin C (Ascorbic acid):

- It is also called anti-scurvy vitamin.
- Ascorbic acid is the most active reducing agent, and a powerful antioxidant.
- It is a vitamin needed to form collagen for blood vessels, cartilage, muscle and bones.
- Vitamin C is also vital to healing of damaged areas of the body.

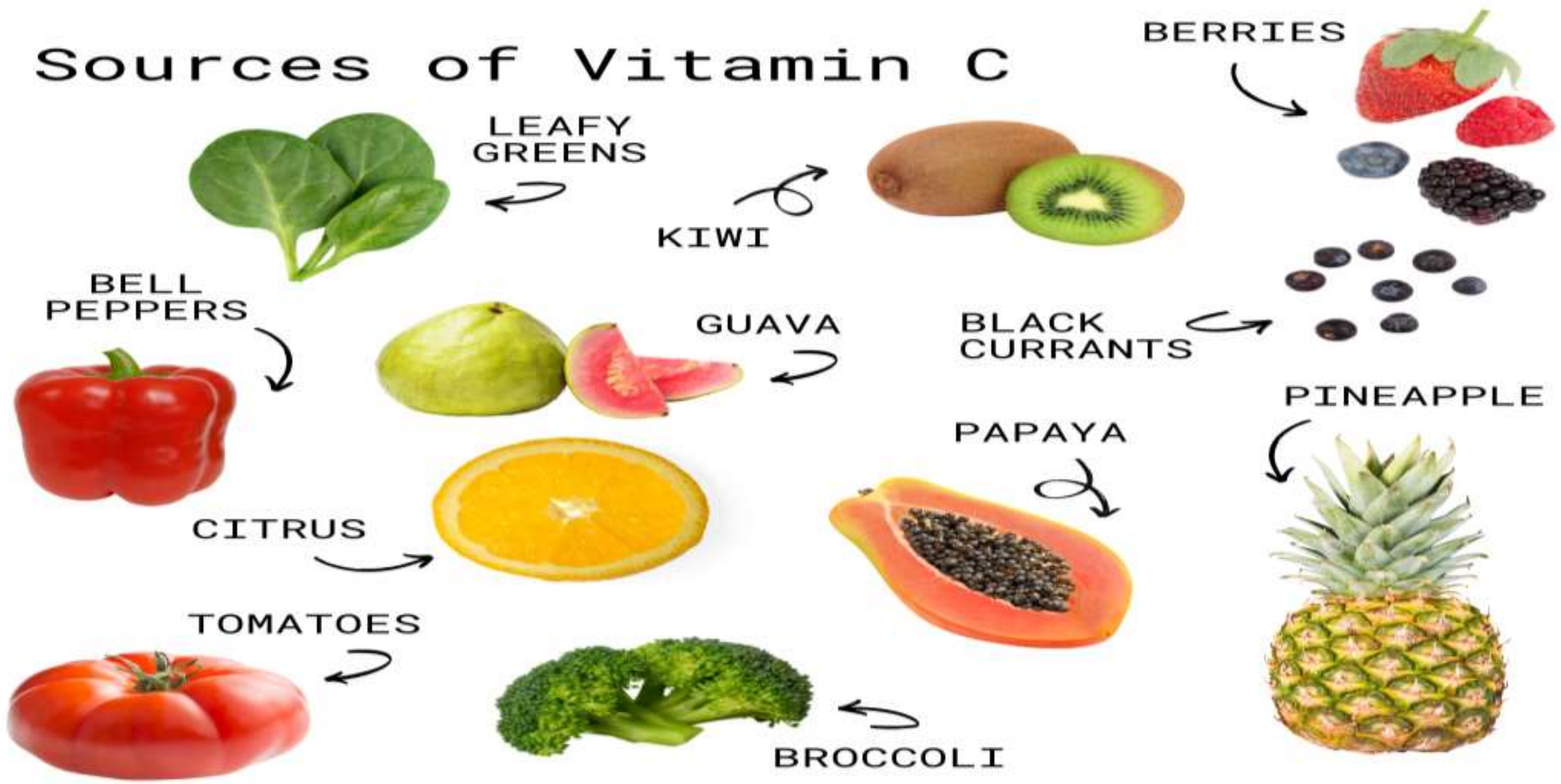


- Necessary for synthesis of collagen due to conversion of proline to hydroxyproline.
- Necessary for maintenance of bones & teeth.
- It stimulates immune function.

Recommended Daily Allowance

90 mg	for men
75 mg	for women

Sources of Vitamin C



Deficiency of Vitamin C:

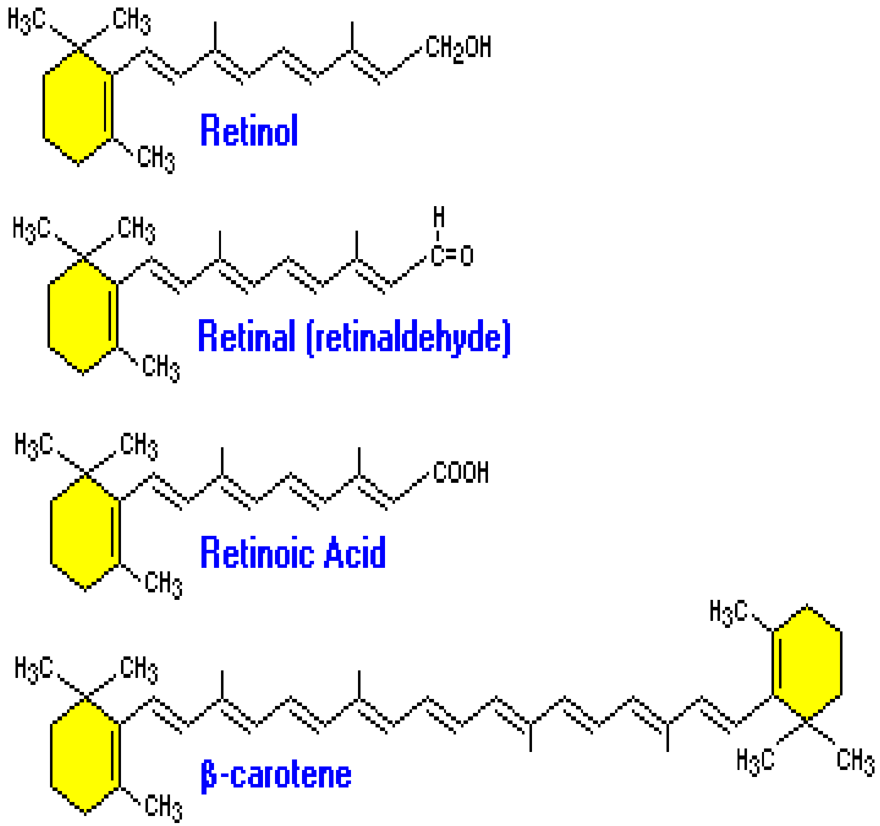
- Scurvy, a disease characterized by:
  - sore and spongy gums, loose teeth.
  - fragile blood vessels, resulting in nose bleeding.
  - swollen joints, petechiae (spot haemorrages).
  - anemia.
- Poor Bone & Dentin formation.
- Impaired immunity.

Swollen joints, petechiae (spot haemorrages), loose of teeth

Fat Soluble Vitamins

1- Vitamin A (Retinol), β- Carotene:

- Family of molecules found in the body in 3 forms; Retinol, Retinal ,Retinoic acid, collectively known as retinoids.
- Retinol, a primary alcohol containing a β-ionone ring with an unsaturated side chain. Named because of its concern with retina of eye & only found in animal foods.
- Plant foods contain β-carotene, which can be oxidatively cleaved in the intestine to yield two molecules of retinal.



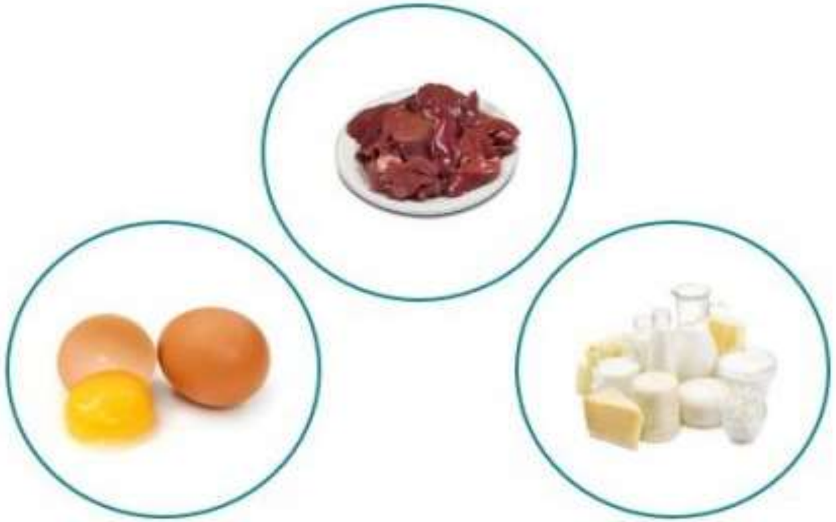
- Retinal is combined with a protein called opsin to give rhodopsin, an essential light absorbing molecule needed for color vision and seeing in dim light.
- Act as an antioxidant, so preventing cardiovascular disease and cancer.

Recommended Daily Allowance

900 µg	for men
700 µg	for women

Sources of Vitamin A

Animal products



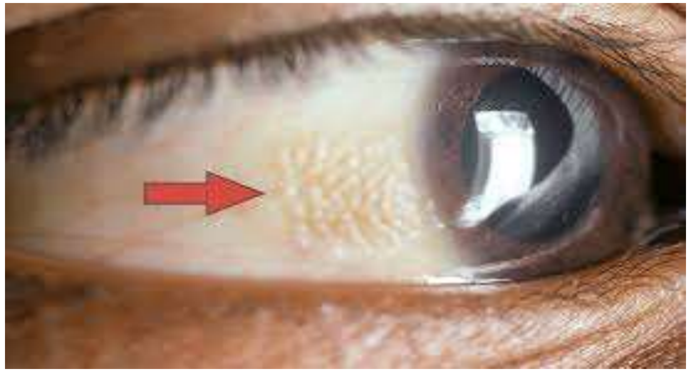
Liver | Egg yolk | Dairy products

Plant products

Orange and red colored fruits and vegetables |
Green leafy vegetables

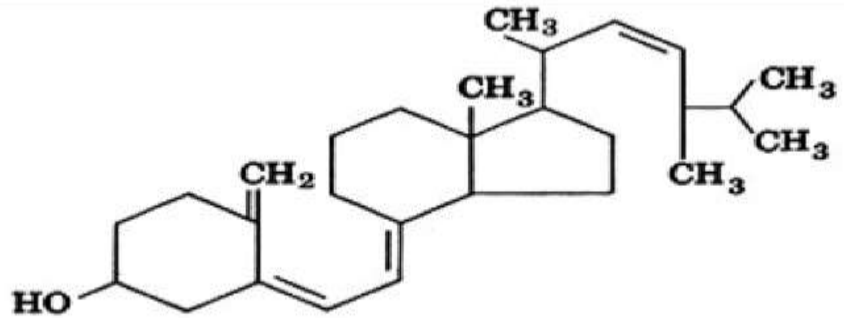
Deficiency of Vitamin A:

- Nyctalopia (night blindness).
- Xerophthalmia, conjunctiva become dry, thick and wrinkled.
- Keratomalacia, cornea becomes soft and may burst.
- Complete blindness (in severe deficiency).

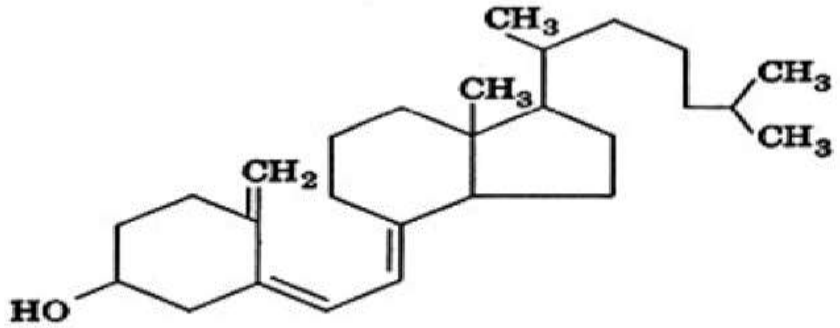


2- Vitamin D (Cholecalciferol & Ergocalciferol):

- The D vitamins are a group of sterols that have a hormone-like function. The active molecule is 1,25(OH)2 D3 (1,25 dihydroxycholecalciferol).
- The sun light activates Provitamin (7-dehydrocholesterol) present under the skin, and convert it into Cholecalciferol.



Vitamin D_2 (ergocalciferol)

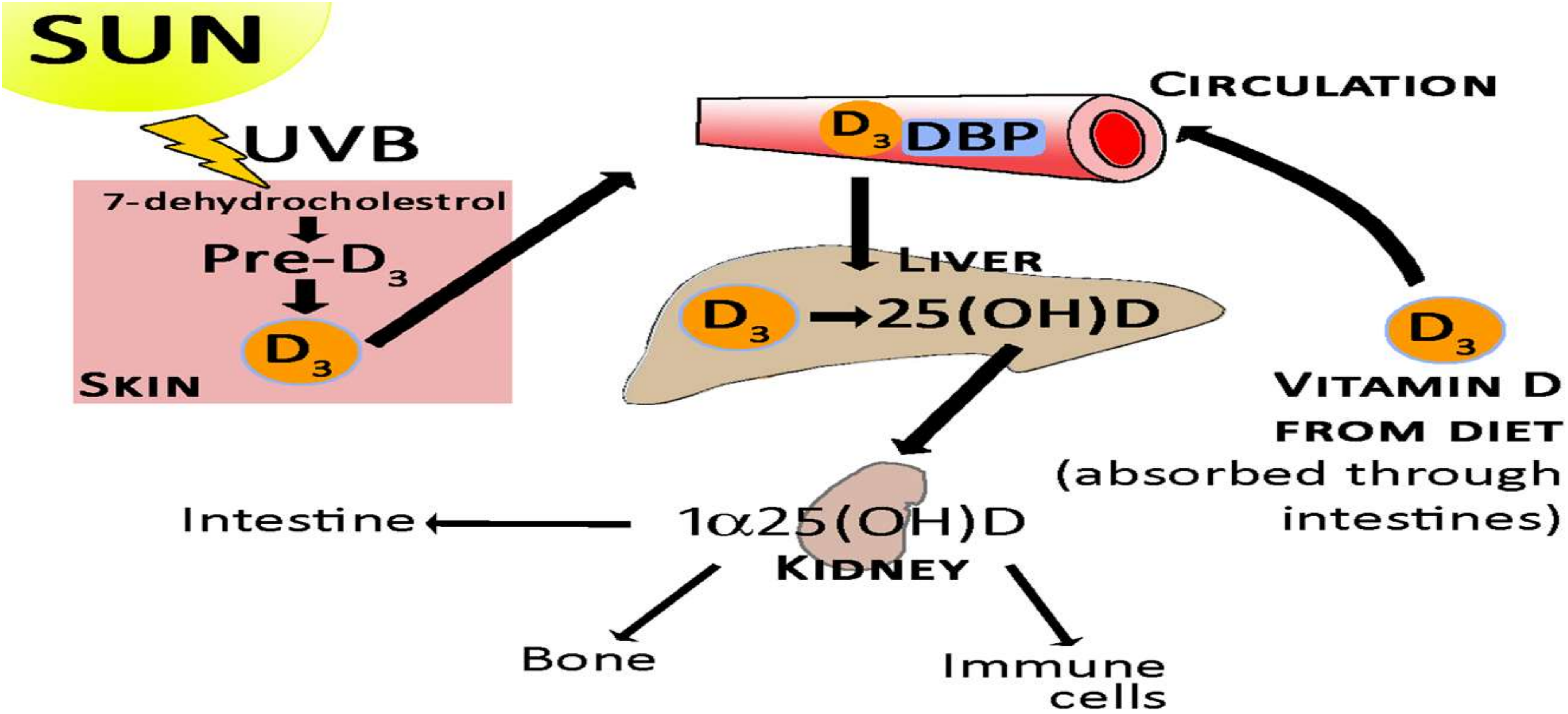


Vitamin D_3 (cholecalciferol)

- Ergocalciferol (vitamin D2), formed by action of ultra violet light on fungi and yeasts, manufactured synthetically for use as vitamin supplement.
- The most prominent actions of vitamin D3 are to regulate the plasma levels of calcium and phosphorus, by

- Increasing absorption of calcium and phosphorus by the intestine.
- Minimizing loss of calcium by the kidney.
- Stimulating bone resorption (osteoclast activity).

How active form of Vitamin D3 produced?



Recommended Daily Allowance

600 IU (15 µg)   for men & women

Sources of Vitamin D

Animal products

Fatty fish | Egg yolk | Cheese | Beef

Plant products

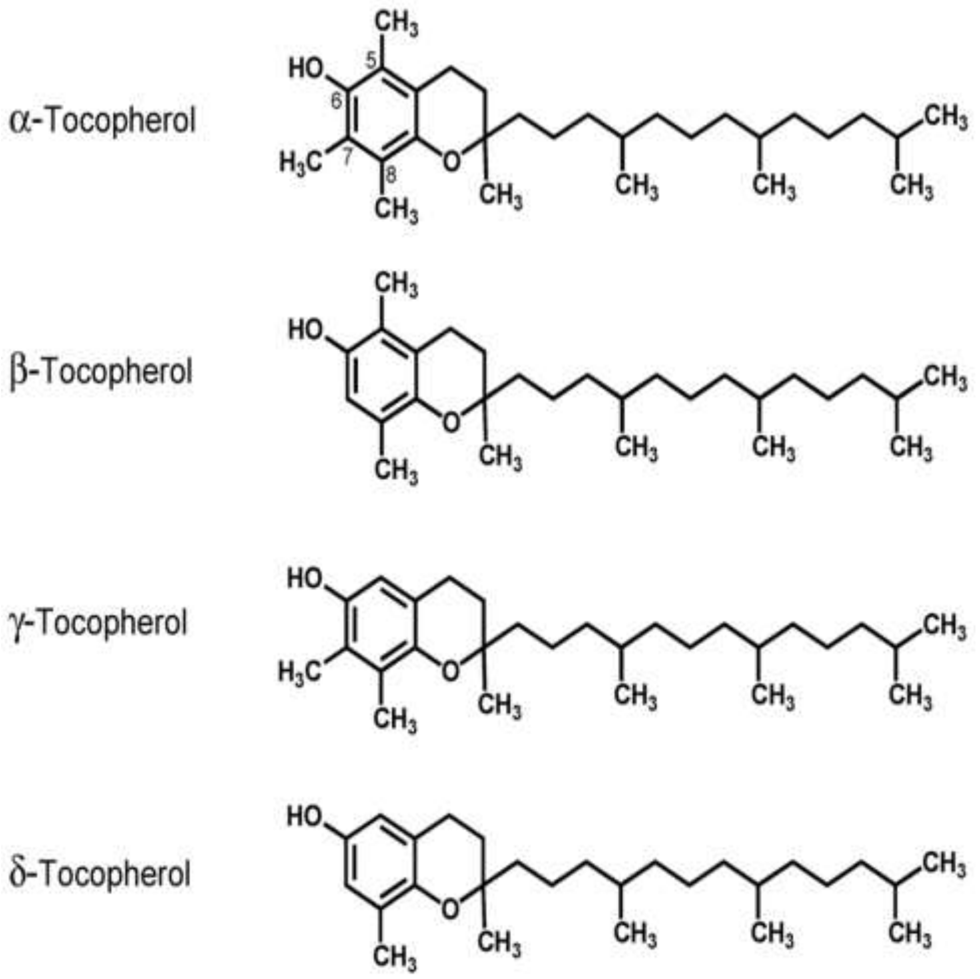
Mushrooms

Deficiency of Vitamin D:

- Demineralization of bone, resulting in rickets in children and osteomalacia in adults.
- Increased risk of bone pain, bone fractures, muscle pain, and muscle weakness.

3- Vitamin E (Tocopherols):

- Vitamin E refers to a group of compounds that include both tocopherols and tocotrienols. They are naturally occurring antioxidant.
- It is also called anti-aging factor.
- Tocopherol α, β, γ, and δ have been obtained from the natural sources.

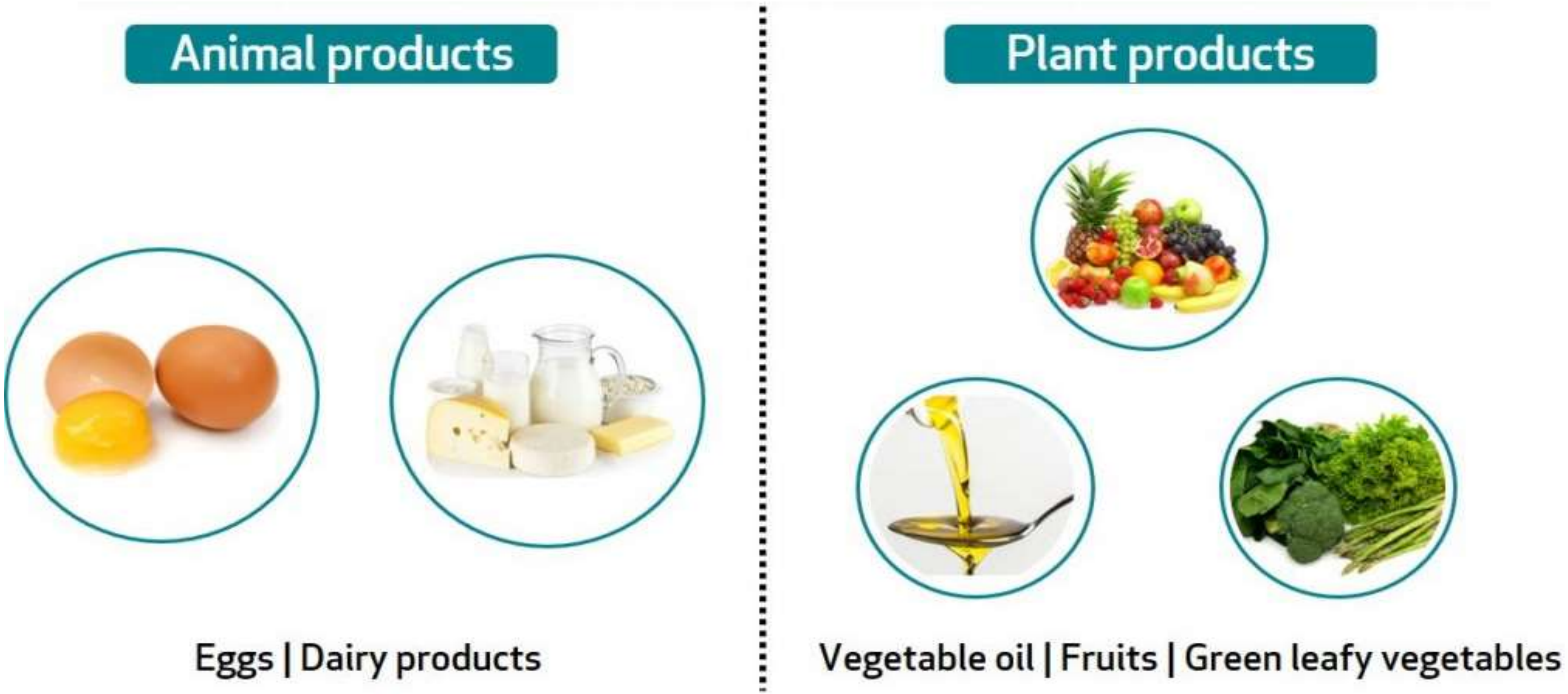


- Through its antioxidant property, vitamin E can:
  - act as a free radical scavenger.
  - protects cell membrane.
  - protects LDL from oxidation.

Recommended Daily Allowance

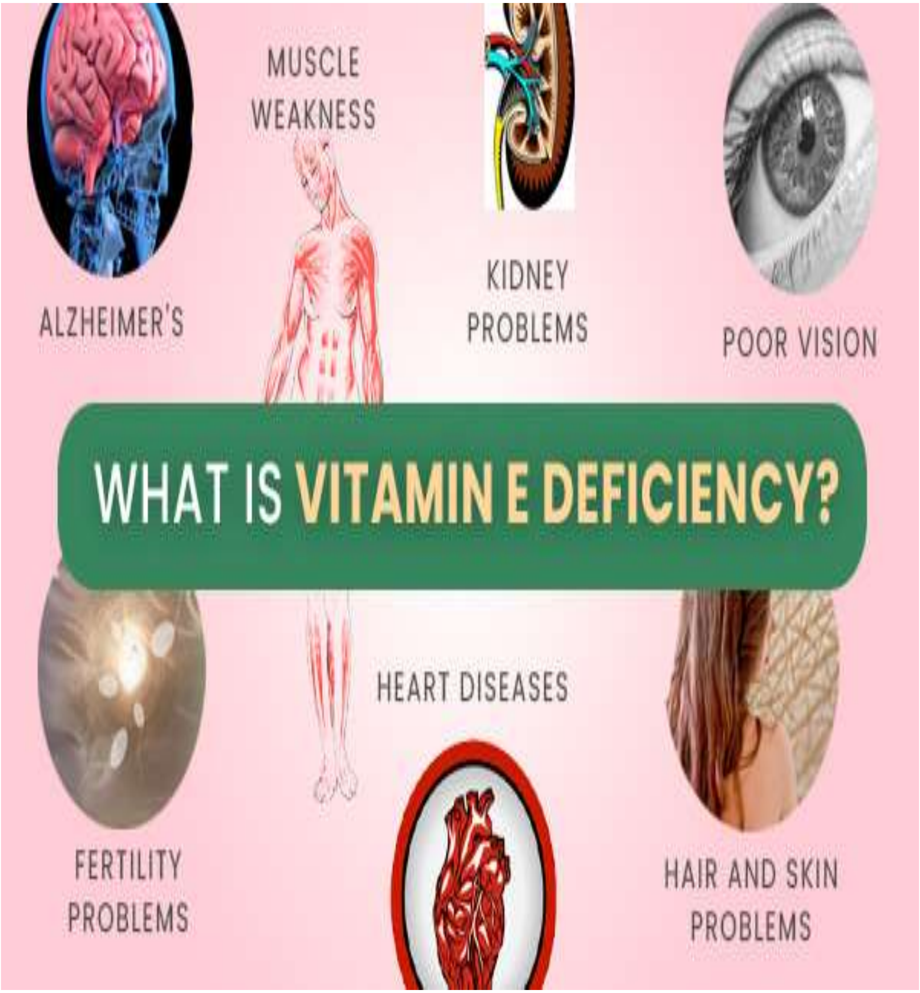
15 mg   for men & women

Sources of Vitamin E



Deficiency of Vitamin E:

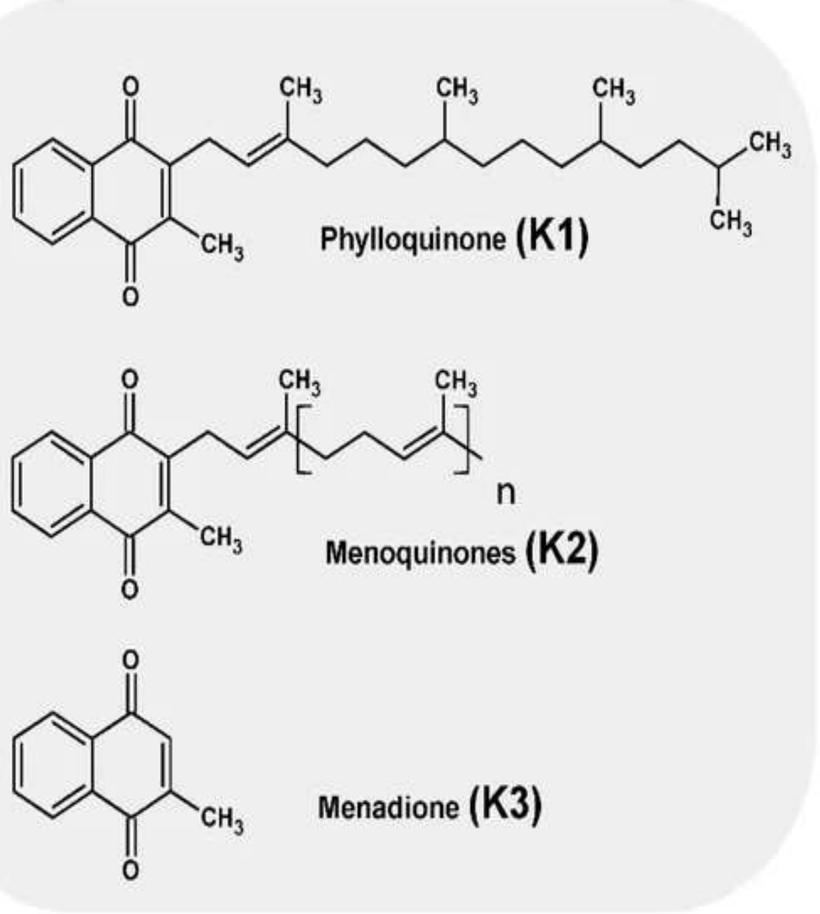
- Appearance of abnormal cellular membranes.
- Neurological problems due to poor nerve conduction.
- Anemia due to oxidative damage to red blood cells.



4- Vitamin K:

➢ Vitamin K exists in several forms:

- Vitamin K1 **(phylloquinone)** is present in plants.
- Vitamin K2 **(menaquinone)** is produced by the intestinal bacterial flora and also found in animals.
- Vitamin K3 **(menadione)** is the synthetic form.



- All the three vitamins (K1, K2, K3) are napthoquinone derivatives.
- Vitamin K is vital for synthesizing clotting factors (II, VII, IX, X), that are necessary for blood clot formation.
- Role in bone metabolism, by it’s requring for the carboxylation of glutamic acid residues of osteocalcin, a Ca^{2+} binding protein present in bone.
- Cardiovascular health, vitamin K helps prevent arterial calcification.

Recommended Daily Allowance

1 µg for men & women

Sources of Vitamin K

Animal products

Dairy products | Meat | Liver

Plant products

Leafy green vegetables | Cabbages | Vegetable oils

Deficiency of Vitamin K:

- Increased tendency to hemorrhages.
- Defective blood clotting.
- Bleeding occurs in nose.
- Prothrombin levels are reduced.

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THANK
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